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Enhanced piezoelectric response of $Bi_4Ti_3O_{12}$-based ceramics through engineered domain configuration and grain size

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A B S T R A C T

High-temperature ferroelectrics are highly desirable for numerous electromechanical devices such as sensors and actuators for high-temperature applications. However, it's challenging to achieve high piezoelectricity and T_C to meet the urgent need for high sensitivity and high temperature. Herein, the piezoelectricity of $Bi_{3.97}Ce_{0.03}Ti_{2.99}W_{0.005}Ta_{0.005}O_{12}$ (BCTWT) ceramics is enhanced by engineering domain configuration and grain size. We have carried out a systematic analysis of the grain size dependent on the electrical performances of BCTWT ceramics. The d_{33} value increases with increasing grain size, a max d_{33} value of 36.6 pC/N is obtained at average grain size of 19.11 μm. Besides, the reduction of domain size, increasing domain wall (DW) density and weak pinning effect induce excellent domain switching performances in coarse-grained BCTWT ceramics, which are mainly responsible for the superior electrical properties. These findings provide a universal approach to achieving high piezoelectricity of BIT-based ceramics by engineering domain configuration and grain size.

Ferroelectric materials, characterized by spontaneous polarization (P_s) that can be switched by an external electrical field, are currently being widely researched for the development of memories, detectors, transducers and piezoelectric sensors [1]. Bismuth layer-structured ferroelectrics (BLSFs) with a representative Aurivillius structure are becoming one of the promising candidates for high-temperature electronic devices due to their high Curie temperature (T_C>500 °C) and excellent temperature stability [2]. Bismuth titanate ($Bi_4Ti_3O_{12}$, BIT), a member of the BLSFs family, has attracted considerable attention in the fields of piezoelectricity and catalysis due to its remarkable ferroelectric, piezoelectric and photocatalytic decomposition performance [3–7]. In terms of the piezoelectricity of BIT ceramics, a number of studies focus on the chemical doping strategies and have achieved significant improvements. For example, Xie et al. have obtained BIT-based ceramics with excellent piezoelectricity (d_{33}~30.5 pC/N) and high-temperature resistivity (ρ~1.24 × 10^7 Ω·cm at 500 °C) by the introduction of Nb^{5+}-Zn^{2+}-Nb^{5+} ion-pair [8]. A high piezoelectric coefficient (d_{33}~31.1 pC/N) and large residual polarization (P_r =12.2 μC/cm^2) with a prominent DC resistance (ρ~5.49 × 10^6 Ω·cm at 500 °C) have been achieved in W/Cr doped BIT-based ceramics by Zhang et al. [9]. Nevertheless, the microscopic origin of the piezoelectric enhancement of BIT ceramics remains ambiguous.

In essence, the piezoelectricity of ferroelectrics originates from two types of contributions, that is, the intrinsic lattice contribution and extrinsic contribution from the domains [10,11]. However, the contribution from the intrinsic lattice is difficult to verify by experimental results due to the lack of single crystal as well as the complicated atomic structure of BIT. Consequently, Tan et al. found that the longitudinal polarization stretching and in-plane polarization rotation contribute most to the piezoelectricity of poled polycrystals as studied by first-principle calculations [12]. On the other hand, due to the peculiar crystal structure of BIT, there is a major component of P_s lies along the *a*-axis [$P_{s(a)}$~50 μC/cm^2] as well as a small component along the *c*-axis [$P_{s(c)}$~4 μC/cm^2] [13]. And these two polarization components, which can be switched independently of each other, can lead to various types of domain structures in BIT ceramics [14]. Using transmission electron microscopy (TEM), Ye et al. have observed five types of ferroelectric DWs in $Bi_4Ti_3O_{12}$ single crystal and revealed the morphology of each type of domain wall [15]. Furthermore, some studies have verified that engineering domain configurations are an effective means to improve the piezoelectric response of BIT ceramics by piezoresponse force microscopy (PFM) and TEM [16–21]. However, chemical etching techniques probably provide a more intuitive picture of the configuration and density of DWs.

Grain size has been verified to have a significant effect on the ferroelectric, dielectric, and piezoelectric properties of some

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ferroelectric ceramics [22–29]. There have been few earlier attempts to study the effect of grain size on the electrical properties of BIT-based ceramics [30,31]. At the same time, the grain size range of these studies is limited to a small interval (about 0.25 μm~5 μm) and the dependence of electrical properties is not significant. The reason for the poor dependency is thought to be related to the control of grain size, while extremely fine and abnormally grown large grains are difficult to obtain.

In the present work, superior piezoelectric response of BIT-based ceramics ($Bi_{3.97}Ce_{0.03}Ti_{2.99}W_{0.005}Ta_{0.005}O_{12}$, BCTWT) is expected through engineered domain configuration. Herein, microwave sintering (MWS) and conventional sintering (CS) were taken to regulate the grain size of BCTWT. It was found that the electrical properties of BCTWT are strongly dependent on grain size owing to the different domain configurations. Significant improvement in piezoelectric and ferroelectric properties (d_{33}~36.6 pC/N, P_r~16.4 μC/cm^2) mainly result from the increased density of $P_{s(a)}-90°$ DWs and $P_{s(c)}-180°$ DWs in coarse-grained (19.11 μm) BCTWT ceramics. As part of the comprehensive size effect study, the present work has also investigated the effect of grain size on the ferroelectric switching and electrical conduction behavior of BCTWT ceramics.

Powder of composition $Bi_{3.97}Ce_{0.03}Ti_{2.99}W_{0.005}Ta_{0.005}O_{12}$ is prepared by solid-state reaction of raw materials Bi_2O_3 (99.999 %), TiO_2 (99.99 %), CeO_2 (99.999 %), WO_3 (99.999 %) and Ta_2O_5 (99.99 %). The appropriate amount of starting powder is ball-milled in ethanol for 12 h After drying, the mixture is calcined at 850 °C for 4 h and subsequently the calcined powder is ball-milled again in ethanol for 12 h After drying, the powder is pelletized with 8 % concentration of polyvinyl alcohol (PVA) and moulded under a uniaxial pressure of 200 MPa. The particles with coarse-grain are sintered by conventional sintering at 1100 °C, 1125 °C and 1150 °C with a heating rate of 3 °C/min for 3 h, while the particles with fine-grain are obtained by microwave sintering at 1000 °C, 1050 °C and 1100 °C with a heating rate of 50 °C/min, holding for 30 min, respectively.

The polished particles are thermally etched at 1050 °C for 20 min and then examined by scanning electron microscopy (S-3400 N, Hitachi, Japan) detector. Chemical etching essays are carried out with HF/HCl/H_2O (volume ratio 1:1:2) and the domain structure of etched surface is then observed by SEM. Meanwhile, the domain amplitudes and phase imagines are obtained by PFM (MFP-3D, Asylum Research, Goleta, CA) scanning.

The temperature dependence of the dielectric constant (ε_r) and dielectric loss (tanδ) are measured by LCR meter (HP4980A, Aligent, USA) under an *ac* electric field of 1 V and at 1 MHz. The *dc* electric resistivity is measured at 10 V within the temperature of 200 °C~700 °C with an interval of 25 °C by the high resistor (Keithley 6487). The ferroelectric loops are measured by ferroelectric analyzer (aix-ACCT, Germany). Poling of pellets is carried out at 200 °C for 20 min under a *dc* field of 12 kV/mm after coating silver on the surface. The d_{33} values of the polarized samples are performed by a quasi-static d_{33} meter (ZJ-6A, Institute of Acoustics, Academia Sinica, China). And the thermal stability of d_{33} values is determined by thermal depolarization. The polarized ceramics are kept at different temperatures from room temperature to 700 °C for 2 h, and their d_{33} values are measured at room temperature.

Scanning electron micrographs of the investigated BCTWT ceramics sintered under different conditions are shown in Fig. 1. Fig. 1 (a-c) show the surface morphology of BCTWT ceramics obtained by MWS. The grain size of the ceramics is controlled to be small due to the rapid heating rate and low sintering temperatures of the MWS process. As the sintering temperature increases of MWS, the matrix grains increase in grain size and their shapes change to the typical platelike grains of BLSFs. From the histogram of size distribution shown in the inset of Fig. 1, the average grain size (lengthwise) of the MWS specimens is 3.54 μm, 8.40 μm and 10.41 μm, respectively. Notably, the CS methods with high sintering temperatures and prolonged holding times can lead to the average grain size of ceramics exceeding 14.5 μm. Furthermore, a few grains growth abnormally at sintering temperatures higher than 1100 °C according to Fig. 1 (d-f). Their grain lengths even >100 μm, and the average grain size even reaches 19.11 μm when the sintering temperature is increased to 1150 °C.

The dependence of the piezoelectric coefficient d_{33} and thermal depoling behaviors on the grain size for BCTWT ceramics are shown in Fig. 2 (a-b), implying that the piezoelectricity of BCTWT ceramics is strongly dependent on the grain size. The piezoelectric coefficient of BCTWT ceramics increases significantly with increasing grain size when the average grain size is below 14.5 μm. However, it is difficult to increase the grain size further when the average grain size exceeds 14.5

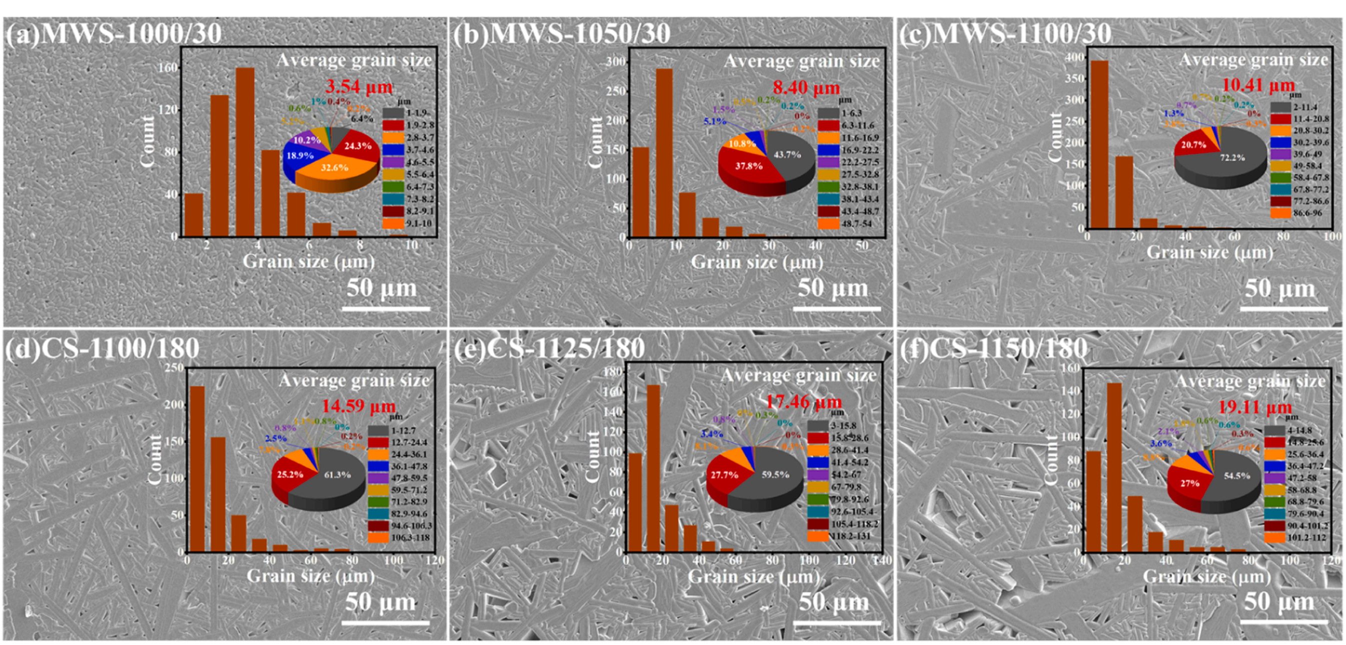

Fig. 1. Surface morphology of BCTWT ceramics with different average grain sizes: (a) 3.54 μm, (b) 8.40 μm, (c) 10.41 μm, (d)14.59 μm, (e) 17.46 μm, and (f) 19.11 μm.

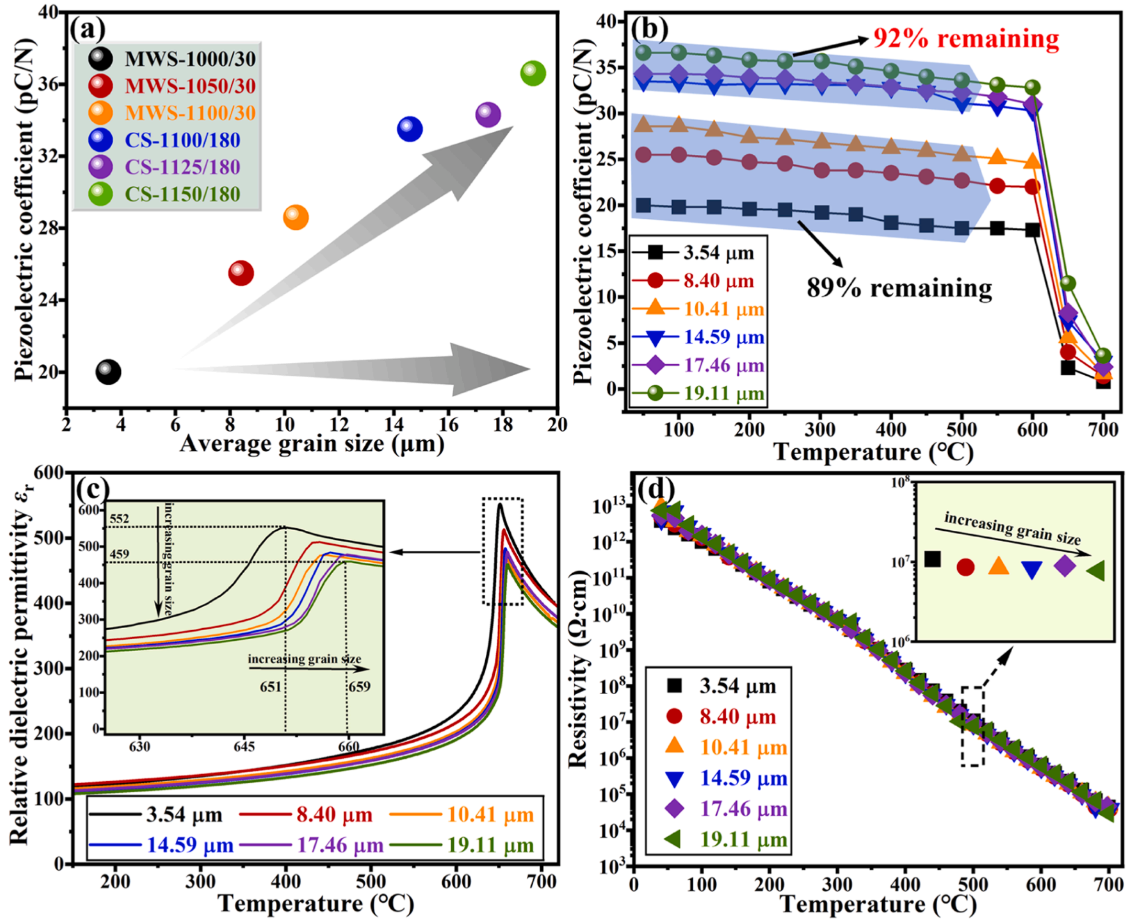

Fig. 2. (a) Piezoelectric coefficient d33, (b) thermal depoling behaviors, (c) relative dielectric permittivity εr and (d) dc resistivity (from room temperature to 700 °C) dependence on grain size of the BCTWT ceramics.

μm and the maximum value of the piezoelectric coefficient remains at 36.6 pC/N with the maximum average grain size of 19.11 μm. Fig. 2(b) shows the thermal depoling behaviors of BCTWT ceramics with different grain sizes. The samples with fine-grain (below 10.41 μm) are less stable to thermal depolarization than the coarse-grained samples (above 14.59 μm), with the coarse-ceramics retaining approximately 92 % of the room temperature d_{33} value after 2 h of depolarization at 500 °C, whereas the fine-ceramics retaining only 89 %. Fig. 2(c) shows the variation of the relative dielectric constant(ε_r) as a function of temperature for BCTWT ceramics with different grain sizes, as measured at 1 MHz. The maximum ε_r gradually decreases from 552 to 459 with increasing in grain size, as shown in the inset of Fig. 2(c). Internal stress may be responsible for the increase of the ε_r in fine-grain BCTWT ceramics [32, 33]. In addition, the peak maximum of ε_r shifts slightly to higher temperature as the grain size increases, suggesting that T_C can be improved by increasing the grain size. As the lattice parameter a/b value decreases in fine-grains, the structure of BCTWT ceramics approaches to the tetragonal structure and the phase transition temperature decreases accordingly [31]. The BCTWT ceramics exhibit excellent high-temperature resistance as shown in Fig. 2(d), which provides a good foundation for their application at high-temperature. It is well known that increasing grain size inevitably leads to the decreasing of boundary density. Accordingly, the DC resistivity at 500 °C of BCTWT ceramics decreases slightly from 1.08×10^7 Ω·cm to 7.72×10^6 Ω·cm during the variations from fine grains to coarse grains. Even though the DC resistivity of coarse-grained ceramics is slightly reduced, the excellent piezoelectric properties and thermal stability are still more favorable for their competition in high-temperature piezoelectric devices.

The *P-E* and *I-E* hysteresis loops measured at 180 °C, 4 Hz and 110 kV/cm of BCTWT ceramics with different grain sizes are shown in Fig. 3 (a-f). It can be observed that all ceramics show the saturated hysteresis loops with apparent ferroelectric switching peaks, indicating sufficient domain reorientation under the external field. Interestingly, the remanent polarizations (P_r) of BCTWT ceramics show a clear positive trend of correlation with the grain size. The P_r value of BCTWT ceramics increases from 3.91 μC/cm^2 to an astonishing 16.40 μC/cm^2 as the grain size increases from 3.54 μm to 19.11 μm. The results suggest that the strong coupling between grain boundaries and domain walls leads to the difficult domain reorientation and inhibition of domain wall motion [26]. This grain boundary effect inevitably results in the reduction of P_r value in fine-grained ceramics due to the reduced achievable domain orientation. In addition, the optimization of P_r value is also partly responsible for the enhanced piezoelectric activities, as follows [34]:

$$d_{33} = 2Q_{11}P_r\varepsilon_{33} \tag{1}$$

where Q_{11} is the longitudinal electrostriction coefficient, P_r is the remnant polarization and ε_{33} is the dielectric constant after DC poling along the direction of polarization. However, for BCTWT ceramics, the coercive field (E_c) exhibits weak grain size dependence. It is clear that coarse-grained ceramics possess excellent comprehensive properties (Table 1), providing a good foundation for high-temperature applications.

It is well known that the piezoelectric properties of piezoelectric ceramics mainly originate from the intrinsic contribution from lattice distortion, which originates from the deformation of the unit cell under the external electric or mechanical fields, and the extrinsic contribution from domain evolution, which is closely related to the domain wall

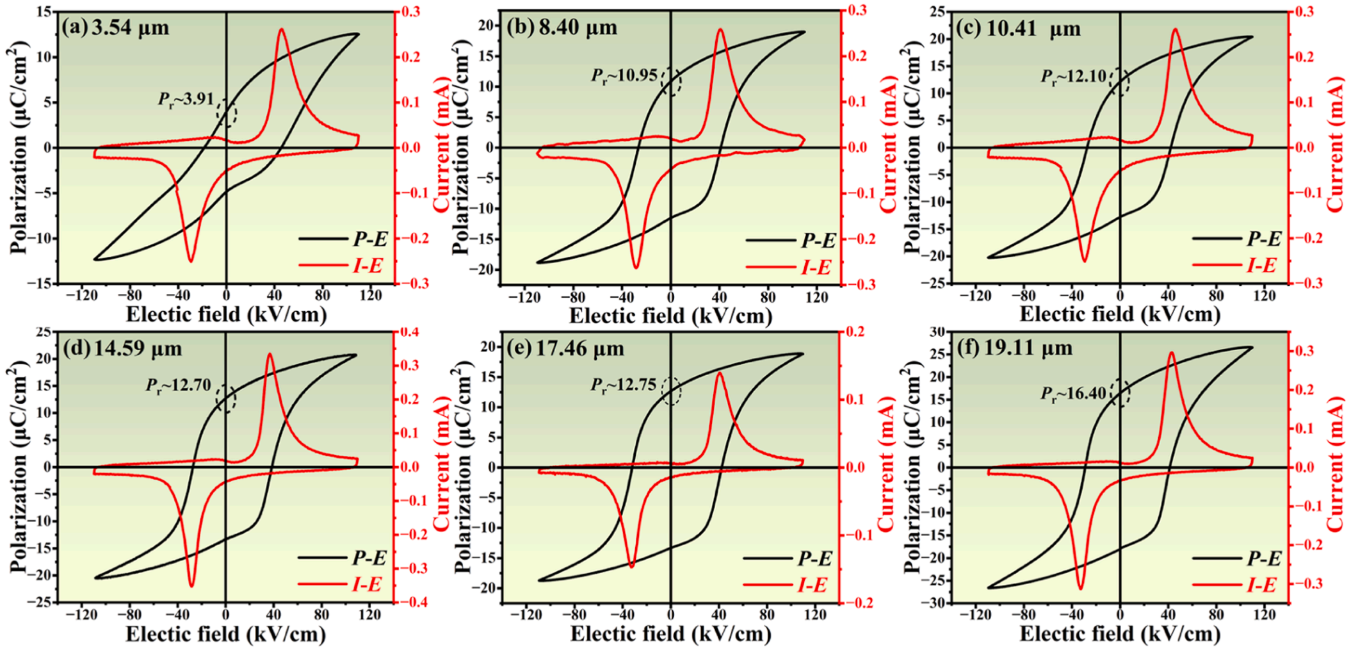

Fig. 3. P-E and I-E hysteresis loops for representative BCTWT ceramics at 180 °C and 4 Hz: (a) 3.54 μm, (b)8.40 μm, (c) 10.41 μm, (d)14.59 μm, (e) 17.46 μm and (f) 19.11 μm.

Table 1
A comparison of the properties for BCTWT ceramics with different gain size.

Average Grain size (μm)	d_{33} (pC/N)	P_r (μC/cm^2)	ε_r @ Cure point	ρ @500 °C (Ω·cm)	T_C (°C)
3.54	20.0	3.91	552	1.08×10^7	651
8.40	25.5	10.95	512	8.53×10^6	654
10.41	28.6	12.10	476	8.38×10^6	656
14.59	33.5	12.70	484	8.42×10^6	657
17.46	34.3	12.75	478	8.92×10^6	659
19.11	36.6	16.40	459	7.71×10^6	659

(DW) density and the domain wall migration, and the domain switching process is significantly affected by the grain size [35]. In the following, EBSD images, XRD patterns and density characteristics of BCTWT ceramics are shown in Figure S1, and the domain configurations of BCTWT ceramics with fine and coarse grains are observed by employing SEM (surfaces morphology by acid etched) and PFM. Furthermore, the evolution of domains after poling process in fine- and coarse-grained ceramics are systematically investigated.

Fig. 4 (a1-a2) show the SEM images of natural and chemically etched domain patterns of unpoled BCTWT ceramics (fine-grained and coarse-grained). Since the positive end of the electric dipole moment of ferroelectrics is etched faster than the negative end, the domains of different polarities are etched by the acid to different degrees resulting in the different domain structures. Here, it is found that the ferroelectric domains mainly present on the lateral planes (*ac*/*bc* plane) of the grains, while there is no domain on the *ab* planes, which is caused by the factor that the spontaneous polarization (P_s) of BIT is mainly along the *a*-axis, P_s vector is restricted to lying close to the *ab* plane [36]. In addition, more domains are observed on the natural surface of the coarse-grained BCTWT ceramics, and the DW density is significantly higher than that of the fine-grained BCTWT ceramics, which would lead to the enhancement of domain wall activity and hence achieve high piezoelectric performance of coarse-grained BCTWT ceramics. To further investigate the effect of grain size on the domain switching behaviors of BCTWT ceramics, the domain configurations of coarse- and fine-grained ceramics before and after polarization are shown in Fig. 4 (b1-c2). It can be seen that the polished surface of fine-grained ceramics still possesses a large number of clearly visible DWs after poling, while there is no trace of DWs on the surface of coarse-grained ceramics. These results demonstrate that the domain switching performance of fine-grained ceramics is much poorer than that of coarse-grained ceramics under an applied electric field. Due to the strong coupling between DWs and grain boundaries, the domain switching in fine-grained ceramics may be pinned by grain boundaries, resulting in the inability to response to electric field, as shown in the schematic in Fig. 4(d).

PFM has become a powerful and versatile tool for probing nanoscale phenomena in ferroelectric materials [37–39]. The strength of the local piezoelectric coupling can be obtained from the amplitude of PFM, while the phase images give an idea of the polarization direction [40]. Herein, PFM is employed to investigate the domain structures of BCTWT ceramics with different grain sizes, as shown in Fig. 4 (e1-f2). The unique layered structure of BIT induces strong anisotropic growth, where the *ab* plane in the typical plate-like grains corresponds to the crystallographic (001) plane (see Fig. 4 (a1, a2)), and the grain boundaries in the PFM images can be labelled as the white dotted lines in Fig. 4 (e1, f1). Accordingly, there are two types of DWs that occur on BCTWT ceramics, where one is (110) DWs intersecting grain boundary or *ab* plane and the other is (001) DWs lying in the *ab* plane or grain boundary [41]. It is clear that fine-grained BCTWT ceramics show predominantly the larger-width domain clumps with (110) DWs, whereas coarse-grained ceramics mainly perform the thin domain stripes with high-density (001) DWs. Convincingly, the reduction of domain size necessarily decreases the domain wall energy due to $D \propto (E_{DW})^{1/2}$(where D and E_{DW} are the domain size and the domain wall energy, respectively) and then improves polarization rotation, which is consistent with the above results from the domain evolution before and after poling. Therefore, it is believed that the superior d_{33} value of coarse-grained BCTWT ceramics is beneficial from the increasing DW density and decreasing domain size.

BCTWT ceramics with average grain size ranging from 3.54 μm to 19.11 μm are fabricated by two different sintering processes: conventional sintering and microwave sintering. The results indicate that the piezoelectricity, dielectric properties and ferroelectricity of BCTWT ceramics show significant grain size dependence. Coarse-grained BCTWT ceramics possess excellent comprehensive properties, with excellent d_{33} value of 36.6 pC/N, T_C of about 659 °C, superior P_r value of 16.40 μC/

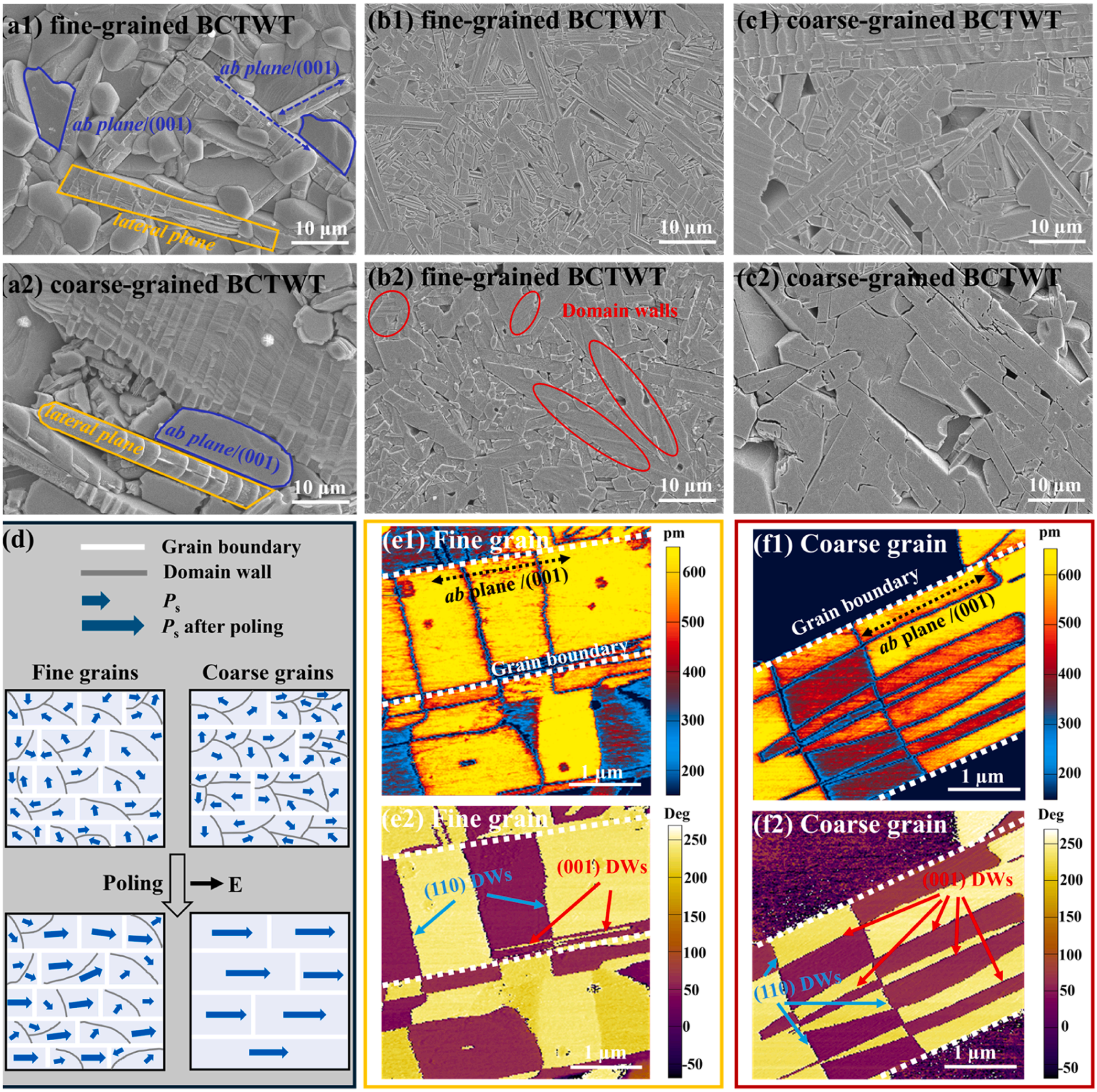

Fig. 4. SEM micrographs of etched surfaces of BCTWT ceramics: (a1-a2) unpolished fine and coarse grained BCTWT ceramics; (b1-b2) polished fine-grained BCTWT ceramics and (c1-c2) polished coarse-grained BCTWT ceramics before and after poling, respectively. (d) Schematics of the domain evolution behaviors for fine grains and coarse grains after poling. (e1-f2) PFM amplitude images and phase images of fine grained and coarse grained BCTWT ceramics.

cm^2 and a modest DC resistivity of 7.72×10^6 Ω·cm obtained in the 19.11 μm samples. The domain configurations of BCTWT ceramics are observed using both acid-etched SEM morphology and PFM images, and the results confirm that the enhanced domain density, excellent domain switching performances and reduced domain size should be mainly responsible for the enhanced piezoelectric performance of coarse-grained BCTWT ceramics. These experimental findings indicate that increasing the grain size of BIT-based ceramics can be an effective modification project without a complex process.

CRediT authorship contribution statement

Wei Shi: Writing – review & editing, Writing – original draft, Software, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yutong Wu:** Methodology. **Hao Tu:** Resources, Methodology. **Shangyi Guan:** Software, Resources. **Jie Xing:** Visualization, Validation, Software, Resources. **Liang Xu:** Methodology. **Hongfei Xu:** Resources, Methodology. **Qiang Chen:** Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.scriptamat.2024.116493.

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